

Introduction to Intelligent Robotics (CC3046)

Project Proposal

Comparative Evaluation of Autonomous Service Robots
in Restaurant Environments

Catarina Fernandes - up202307632

Leonor Baía - up202304808

Pedro Ferreira - up202208344

Goals of the work

The goal of this project is to study how environmental knowledge influences the performance of autonomous waiter robots in restaurant service tasks. To this end, we will analyse two different navigation approaches: a robot that operates with prior knowledge of the restaurant layout, using map-based path planning, and a robot that starts without a map and incrementally builds a representation of the environment and an estimation of its own position, while navigating.

The service task considered in this work consists of detecting or receiving a request from a specific table, moving safely from the base to that table, and returning to the starting point after completing the service. In order to assess the behaviour of both robots under different conditions, the experiments will be carried out in both static and dynamic restaurant environments.

Through this work, we aim to evaluate how each navigation strategy affects efficiency, safety, adaptability, and overall task completion in a realistic indoor service scenario. In particular, the project seeks to achieve the following outcomes:

- Development of a restaurant service simulation environment for autonomous waiter robots;
- Implementation of two distinct navigation approaches, with and without prior environment knowledge;
- Evaluation of both approaches in static and dynamic restaurant environments;
- Comparative analysis of task success, time efficiency, collision avoidance, and adaptability to environmental changes.

Proposed Approach

This project will be developed in the Webots simulation environment, where a mobile robot navigates in a structured indoor restaurant scenario.

The robotic platform used will be the **e-puck robot**, equipped with:

- **LiDAR sensor** for environment perception;
- **Wheel encoders** for odometry estimation;
- (Optional) **Compass** for orientation refinement.

The robot must navigate between a base location and target tables while avoiding obstacles and completing service tasks.

Environment

The restaurant environment will be designed as a structured indoor layout within the simulation, including:

- **Static obstacles**, such as tables, chairs, and walls;
- **Navigable corridors** between tables, allowing safe robot movement;
- A **fixed base station**, serving as both the starting and ending point for all tasks;
- **Dynamic obstacles**, represented by moving agents simulating people (in the dynamic scenario).

The environment will be developed using a hybrid approach combining simulation efficiency and visual realism. Initially, simplified geometric representations will be used to ensure reliable robot perception and stable collision handling.

As a potential extension, a more detailed 3D model of the restaurant may be created using external design tools and imported into the simulator to enhance visual quality. In this case, simplified geometric representations will still be used as collision models (bounding objects), while detailed meshes are retained for visualization only. This separation ensures computational efficiency while maintaining a realistic scenario.

Action Space

The robot will operate using a **discrete action space**, consisting of:

- Move forward;
- Move backward;
- Turn left;
- Turn right;
- Stop.

Algorithms

Approach A — Map-Based Navigation

This approach represents a strong baseline with full environmental knowledge.

- **Global planner:** A*
- **Localization:** assumed known or odometry-based
- **Control:** rule-based navigation

Approach B — SLAM-Based Navigation

The robot incrementally builds a map while simultaneously estimating its pose.

- **Mapping:** occupancy grid
- **Localization:** odometry + LiDAR-based estimation
- **SLAM pipeline:**
 - sensor data acquisition
 - map update
 - pose estimation
- **Path Planning:** When a table requests service, the robot plans a path to the target using A* over the **currently constructed partial map**. As the map is incomplete, the planned path may be updated dynamically as new obstacles and free space are discovered. This allows the robot to navigate toward the goal while continuously exploring and adapting to the environment.

Proposed Performance Evaluation Experiments

Experiment 1 — Static Environment with Known Map

In this experiment, the robot operates in a static restaurant environment with full prior knowledge of the layout.

- Navigation method: map-based navigation with A* (global path planning)
- Map: fully known in advance
- Obstacles: static only
- Task: navigate from base to a target table and return

This experiment establishes a performance baseline, representing an ideal scenario where the environment is completely known.

Experiment 2 — Static Environment with Unknown Map

In this scenario, the robot does not have prior knowledge of the environment and must build a map while performing the task.

- Navigation method: SLAM-based mapping with incremental mapping and A* planning over the currently known map
- Map: incrementally constructed during execution
- Obstacles: static only
- Task: navigate to the target table while exploring and return to base

This experiment evaluates the robot's ability to perform service tasks while simultaneously exploring and mapping the environment.

Experiment 3 — Dynamic Environment with Known Map

This experiment introduces dynamic elements into the environment while maintaining prior knowledge of the layout.

- Navigation method: map-based navigation with A* and dynamic path replanning
- Map: known
- Obstacles: static + moving agents (simulated people)
- Task: reach the table and return while avoiding moving obstacles

This scenario tests the robustness of the map-based approach under real-world-like disturbances.

Experiment 4 — Dynamic Environment with Unknown Map

This is the most challenging scenario, where the robot must both explore and react to a changing environment.

- Navigation method: SLAM-based navigation with incremental mapping, A* planning over the partial map, and dynamic path updating
- Map: incrementally built
- Obstacles: static + dynamic
- Task: complete delivery and return while mapping and adapting

This experiment evaluates adaptability, robustness, and learning capability in complex environments.

Across all experiments, multiple runs will be performed to ensure statistical relevance and reproducibility of results.

Performance Metrics

To quantitatively evaluate the performance of the proposed navigation approaches, the following metrics will be used:

1. Delivery Time

Time required for the robot to move from the base station to the target table.

- Measures efficiency of forward navigation
- Lower values indicate better performance

2. Return Time

Time required for the robot to return from the table to the base.

- Important for evaluating consistency and path reuse
- In map-based approaches, expected to decrease due to known paths

3. Total Mission Time (delivery time + return time)

Total time to complete the full task Base → Table → Base.

- Primary performance indicator
- Combines delivery and return efficiency

4. Total Mission Distance

Total distance travelled during the mission.

- Measures path optimality
- Helps identify inefficient or exploratory behaviour

5. Number of Collisions per Execution

Total number of collisions with obstacles during a mission.

- Measures safety and robustness
- Critical in dynamic environments

6. Average Speed

Average velocity of the robot during the mission.

- Indicates motion efficiency and smoothness
- Can reveal stop-and-go behaviour

7. Wrong Table Rate

Percentage of executions where the robot fails to reach the correct target table.

- Measures task accuracy
- Especially relevant in exploration-based approaches

8. Success Rate

Percentage of missions successfully completed (delivery and return).

- Measures reliability of the system
- Includes failures due to navigation or collisions

9. Path Efficiency Ratio

Ratio between the optimal path length and the actual path taken.

- Value close to 1 indicates near-optimal navigation
- Useful to compare A* vs exploration-based approaches

10. Exploration Overhead (for SLAM-based approach)

Extra distance or time spent exploring unknown areas.

- Highlights cost of lack of prior knowledge
- Important for comparing efficiency trade-offs

All metrics will be averaged across multiple runs for each experimental setup to ensure statistical significance.

Weekly Work Plan

Week 1 (16/04 - 23/04):

1. Select the restaurant layout design tool and begin modelling the environment;
2. Set up the Webots world and import the e-puck robot;
3. Define the base station, table positions, corridors, and main obstacles;
4. Start organizing the structure of the scientific report.

Week 2 (23/04 - 30/04):

1. Configure the robot sensors and actuators in Webots;
2. Implement the table request system and target identification logic;
3. Perform initial tests to validate robot movement inside the environment;
4. Document the simulation setup and robot configuration for the report.

Week 3 (30/04 - 07/05):

1. Implement the map-based robot with prior knowledge of the environment;
2. Implement and test the A* path planning algorithm for the known-map robot and integrate local obstacle avoidance and motion control;
3. Test the robot in the static restaurant environment;
4. Record initial observations and implementation details.

Week 4 (07/05 - 14/05):

1. Implement the robot without prior map knowledge;
2. Develop the SLAM-based mapping and localization component;
3. Implement and test A* path planning over the incrementally built map;
4. Integrate navigation and path-updating logic for the unknown-map approach;
5. Perform preliminary tests in the static environment;

Objectives for Checkpoint - 14th May

- Completed restaurant environment in Webots;
- Implemented request/call mechanism for the tables;
- First version of the known-map navigation approach and of the unknown-map robot;
- Preliminary tests in the static environment.

Week 5 (14/05 - 21/05):

1. Extend the environment to include dynamic obstacles, such as moving agents;
2. Adapt the known-map robot to dynamic conditions with replanning;

3. Test both robots in static and dynamic environments;
4. Refine the implementation of both navigation approaches.

Week 6 (21/05 - 28/05):

1. Refine behaviour and fix implementation issues;
2. Run the full set of experiments for the four scenarios;
3. Collect performance data for all defined metrics and organize the results into tables and graphs;
4. Write the experimental results section of the report.

Week 7 (28/05 - 04/06):

1. Perform the comparative analysis of both approaches;
2. Finalize the report and prepare the presentation;
3. Record demonstration videos of the experiments;